

BANARAS HINDU UNIVERSITY

Under Graduate Semester III Mid-Term Examination, 2025-26 (NEP VAC)

DEPARTMENT OF STATISTICS**Paper Code: STAMV-32 : Applied Statistics Theory VAC (Mid-Term)**

Time: 1 Hour

Maximum Marks: 20

Instructions: Attempt any FIVE questions. All questions carry equal marks (4 marks each).

Question 1. Plot Normal density function $f(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$ in MS Excel for $x \in [-3, 3]$ and interpret findings. [4]

Question 2. Estimate parameter λ and fit Poisson distribution to website server errors per hour data. [4]

Question 3. Estimate parameters μ and σ and fit Normal distribution to 120 student test marks. [4]

Question 4. Explain steps to compute Chi-square goodness of fit statistic in MS Excel. [4]

Question 5. Compare Binomial vs Poisson models for sparse counts data. [4]